CSE400 Lecture 4 Scribe

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Imagine you are expert scribe generator. Make a lecture scribe suitable for exam preparation for CSE400 Lecture 4: Joint Probability and Conditional Probability. When you are writing the scribe, do reasoning step by step through definitions, assumptions, proofs, derivations, and worked example solutions exactly as presented in the provided context. Include this reasoning in the scribe so that the logical flow is clear for revision. Use only the provided lecture slides and relevant textbook excerpts. Don't mention any material, explanations or examples that are not present in the context.
Organize the scribe clearly in a formal academic style.

Below is a **formal, exam-oriented lecture scribe** for **CSE400 – Lecture 4: Joint Probability and Conditional Probability**, written **strictly from the provided lecture slides** and organized for **revision and answer-writing**.
All definitions, logical steps, statements, and examples follow the **exact order and scope of the slides**, with **explicit reasoning included** to make the flow clear for exams.
No external material or extra examples are introduced.
(Reference: *Lecture 4 slides, pages as indicated*)

CSE400 – Fundamentals of Probability in Computing

Lecture 4: Joint Probability and Conditional Probability

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1. Introduction to Probability Theory

This lecture builds upon the foundational elements of probability theory, which are necessary before defining **joint probability** and **conditional probability**. The lecture first formalizes experiments, outcomes, events, and the probability axioms.

2. Experiments, Sample Space, and Events

2.1 Experiment

Definition:
An **experiment (E)** is a procedure that produces some result.

Example:

- Tossing a coin five times is denoted as experiment E_5.

The reasoning here is that probability theory begins with a repeatable process whose outcomes may vary, even if the procedure itself is fixed.

2.2 Outcome

Definition:
An **outcome (ξ)** is a possible result of an experiment.

Example:

- One possible outcome of E_5 is

$$\xi_1 = \text{HHTHT}$$

Each outcome represents one exact realization of the experiment.

2.3 Event

Definition:
An **event** is a specific set of outcomes of an experiment.

Example:

- Let C be the event consisting of **all outcomes with an even number of heads** in E_5.

Thus, an event is a collection (subset) of outcomes, not a single outcome.

2.4 Sample Space

Definition:

The **sample space (S)** is the set of **all possible distinct outcomes** of an experiment.

Key properties:

- Outcomes are **mutually exclusive**: only one outcome can occur at a time.
- Outcomes are **collectively exhaustive**: no possible result lies outside the sample space.

Example:

- For a single coin flip:
  - Heads and tails are mutually exclusive.
  - No other outcome is possible.

2.5 Types of Sample Spaces

The lecture classifies sample spaces as:

- **Discrete**
- **Countably infinite**
- **Continuous**

Examples discussed:

- Flipping a fair coin once
- Rolling a six-sided die
- Rolling two dice
- Flipping a coin until tails occurs
- Random number generation in the interval $[0, 1)$

These examples show how different experiments lead to different structures of S.

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3. Probability and Axioms of Probability

3.1 Definition of Probability

Probability is a numerical measure of the likelihood of an event.

It is defined as a function that assigns a real number to each event.

3.2 Probability Axioms

The axioms are taken as **self-evident truths** and require no proof.

Let A be any event:

1. **Non-negativity and normalization**

$$0 \leqslant \Pr(A) \leqslant 1$$

2. **Certain event**

$$\Pr(S) = 1$$

3. **Additivity for mutually exclusive events**
   - If $A \cap B = \varnothing$, then

$$\Pr(A \cup B) = \Pr(A) + \Pr(B)$$

4. **Countable additivity**
   - For mutually exclusive events A_i:

$$\Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \Pr(A_i)$$

These axioms form the mathematical foundation on which joint and conditional probabilities are defined.

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4. Corollaries from Probability Axioms

Corollary 2.1

For a **finite** number M of mutually exclusive events:

$$\Pr\left(\bigcup_{i=1}^{M} A_i\right) = \sum_{i=1}^{M} \Pr(A_i)$$

Reasoning:

- When the sample space is finite, the infinite additivity axiom reduces to a finite sum.

The lecture emphasizes that **Axiom 3 is necessary** to handle infinite sample spaces.

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5. Propositions from Probability Axioms

Proposition 2.1 (Complement Rule)

$$\Pr(A^c) = 1 - \Pr(A)$$

Reasoning:

- Event A and its complement A^c together form the sample space.
- Since $\Pr(S) = 1$, subtracting $\Pr(A)$ yields $\Pr(A^c)$.

Proposition 2.2 (Monotonicity)

If $A \subset B$, then:

$$\Pr(A) \leqslant \Pr(B)$$

Reasoning:

- Event B includes all outcomes of A and possibly more.
- Therefore, its probability cannot be smaller.

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6. Joint Probability

6.1 Motivation

Joint probability is required when **two or more events are considered simultaneously**.

Instead of asking:

- "What is the probability of event A?"
  we ask:
- "What is the probability that **both** A and B occur?"

6.2 Definition

The **joint probability** of events A and B is:

$$\Pr(A \cap B)$$

This measures the likelihood that **both events occur at the same time**.

6.3 Examples (from slides)

- **Card Deck Example**
- **Costume Party Example**

In each case:

- Individual events are first identified.
- The intersection of those events represents their joint occurrence.
- Probability is assigned to the intersection.

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7. Conditional Probability

7.1 Motivation

Conditional probability answers the question:

> *What is the probability of event A, given that event B has already occurred?*

This reflects **updated knowledge**.

7.2 Definition

The **conditional probability** of A given B is:

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}, \quad \Pr(B) > 0$$

7.3 Logical Reasoning Behind the Formula

- Once B has occurred, the **effective sample space** is reduced to B.
- The favorable outcomes are those in $A \cap B$.
- Dividing by $\Pr(B)$ normalizes the probability within the reduced space.

7.4 Examples (from slides)

- **Cards without replacement**
- **Game of Poker**
- **The Missing Key**

Each example:

1. Identifies the given condition.
2. Restricts the sample space accordingly.
3. Computes probability using the conditional probability formula.

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8. Summary of Key Exam Points

- Clearly distinguish **outcomes**, **events**, and **sample space**.
- State probability axioms **before** using them in derivations.
- Joint probability refers to **intersection** of events.
- Conditional probability is a **ratio of joint probability to the conditioning event**.
- Always check that $\Pr(B) > 0$ before applying conditional probability.

End of Lecture 4 Scribe

(Prepared strictly from Lecture 4 slides for exam preparation)

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Below is a **formal, exam-ready lecture scribe** for **CSE400 – Lecture 4: Joint Probability and Conditional Probability**, written **strictly from the provided lecture**

slides.
The presentation follows the **same order, definitions, statements, and logical development as in the slides**, with **step-by-step reasoning explicitly written out** to support revision and exam answer construction.
No external material, interpretations, or additional examples are included.

CSE400 – Fundamentals of Probability in Computing

Lecture 4: Joint Probability and Conditional Probability

(Based exclusively on Lecture 4 slides)

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1. Introduction to Probability Theory

The purpose of this part of the lecture is to establish the basic probabilistic framework required before introducing **joint probability** and **conditional probability**. This includes formal definitions of experiments, outcomes, events, sample spaces, and probability axioms.

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2. Experiments, Sample Space, and Events

2.1 Experiment

Definition:
An **experiment (E)** is a procedure that we perform which produces some result.

Example (as given):

- Tossing a coin five times, denoted by E_5.

Reasoning:
Probability theory begins with an experiment because probability measures are assigned only after a repeatable process is clearly defined.

2.2 Outcome

Definition:
An **outcome (ξ)** is a possible result of an experiment.

Example:

- One possible outcome of E_5 is

$$\xi_1 = \text{HHTHT}$$

Reasoning:
An outcome represents one exact realization of the experiment. Each experiment may have multiple distinct outcomes.

2.3 Event

Definition:
An **event** (denoted by any letter) is a **certain set of outcomes** of an experiment.

Example:

- Let C be an event in experiment E_5 such that

$$C = \{\text{all outcomes consisting of an even number of heads}\}$$

Reasoning:
An event is not a single outcome but a collection of outcomes that satisfy a specified condition.

2.4 Sample Space

Definition:
The **sample space (S)** is the collection or set of **all possible distinct outcomes** of an experiment.

Key properties (as stated):

- **Mutually exclusive:** Only one outcome can occur at a time.
- **Collectively exhaustive:** No outcome lies outside the sample space.

Example:

- For flipping a coin:

- Heads and tails are mutually exclusive.
- No outcome other than heads or tails is possible.

2.5 Types of Sample Spaces

The lecture classifies sample spaces as:

- **Discrete**
- **Countably infinite**
- **Continuous**

Examples listed in the slides:

- Flipping a fair coin once
- Rolling a cubical die with numbered faces
- Rolling two dice
- Flipping a coin until tails occurs
- Random number generator with interval $[0, 1)$

Reasoning:
Different experiments lead to different structures of sample spaces, which affects how probabilities are assigned.

---

3. Probability and Axioms of Probability

3.1 Definition of Probability

Probability is defined as:

- A measure of the likelihood of various events, **or**
- A function of an event that produces a numerical quantity measuring how likely that event is.

3.2 Probability Axioms

The axioms are taken to be **self-evident** and require no proof.

Let A be any event:

1. **Boundedness**

$$0 \leqslant \Pr(A) \leqslant 1$$

2. **Certain event**

$$\Pr(S) = 1$$

3. **Additivity for mutually exclusive events**
   - If $A \cap B = \varnothing$, then

$$\Pr(A \cup B) = \Pr(A) + \Pr(B)$$

4. **Countable additivity**
   - For an infinite number of mutually exclusive events A_i:

$$\Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \Pr(A_i)$$

Reasoning:
These axioms define the mathematical behavior of probability and serve as the basis for all subsequent results.

---

4. Corollary from Probability Axioms

Corollary 2.1

For a **finite** number M of mutually exclusive events:

$$\Pr\left(\bigcup_{i=1}^{M} A_i\right) = \sum_{i=1}^{M} \Pr(A_i)$$

Explanation (as per slides):

- Axiom 3 reduces to this corollary when the sample space is finite.
- Axiom 3 is necessary for handling infinite sample spaces.

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5. Propositions from Probability Axioms

Proposition 2.1 (Complement Rule)

$$\Pr(A^c) = 1 - \Pr(A)$$

Reasoning:

- $A \cup A^c = S$

- Since $\Pr(S) = 1$, subtracting $\Pr(A)$ yields $\Pr(A^c)$.

Proposition 2.2 (Monotonicity)

If $A \subset B$, then:

$$\Pr(A) \leqslant \Pr(B)$$

Reasoning:
Event B contains all outcomes of A and possibly more, so its probability cannot be smaller.

---

6. Joint Probability

6.1 Motivation

Joint probability is required when **two events are considered simultaneously**. Instead of evaluating single events independently, we evaluate their **intersection**.

6.2 Definition

The **joint probability** of events A and B is:

$$\Pr(A \cap B)$$

This represents the probability that **both events occur together**.

6.3 Examples Mentioned in Slides

- **Card Deck Example**
- **Costume Party Example**

In each case:

1. Events are defined individually.
2. Their intersection represents simultaneous occurrence.
3. Probability is assigned to the intersection.

---

7. Conditional Probability

7.1 Motivation

Conditional probability quantifies uncertainty **after additional information is known**.

The question addressed is:

> What is the probability of event A given that event B has already occurred?

7.2 Definition

For $\Pr(B) > 0$, the **conditional probability** of A given B is:

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

7.3 Step-by-Step Reasoning

1. The occurrence of B restricts the sample space to B.
2. Within this reduced sample space, the favorable outcomes are those in $A \cap B$.
3. Dividing by $\Pr(B)$ normalizes the probability over the restricted space.

7.4 Examples Listed in Slides

- Cards without replacement
- Game of Poker
- The Missing Key

Each example follows the same logical structure:

1. Identify the condition.
2. Restrict the sample space.
3. Apply the conditional probability formula.

---

8. Exam-Focused Summary

- Experiments define the probabilistic process.

- Outcomes form the elements of the sample space.
- Events are sets of outcomes.
- Probability axioms are the foundation for all derivations.
- Joint probability corresponds to event intersections.
- Conditional probability is a normalized joint probability under a given condition.
- Always ensure $\Pr(B) > 0$ before computing $\Pr(A \mid B)$.

End of Lecture 4 Scribe

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The exposition follows the **same structure, definitions, assumptions, propositions, and logical flow as in the slides**, with **step-by-step reasoning explicitly written** to aid exam revision.
No material, intuition, or examples beyond the given context are introduced.

CSE400 – Fundamentals of Probability in Computing

Lecture 4: Joint Probability and Conditional Probability

(Based exclusively on Lecture 4 slides)

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1. Introduction to Probability Theory

This lecture continues the formal development of probability theory and prepares the foundation required to define **joint probability** and **conditional probability**. The lecture begins with precise definitions of experiments, outcomes, events, and sample spaces, followed by axioms and their consequences.

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2. Experiments, Outcomes, and Events

2.1 Experiment

Definition:
An **experiment (E)** is a procedure that we perform which produces some result.

Example (from slides):

- Tossing a coin five times, denoted as E_5.

Logical role:
An experiment specifies the probabilistic setting. Probabilities are always defined with respect to an experiment.

2.2 Outcome

Definition:
An **outcome (ξ)** is a possible result of an experiment.

Example:

- One possible outcome of E_5 is

$$\xi_1 = \text{HHTHT}$$

Logical role:
Each outcome represents one complete realization of the experiment.

2.3 Event

Definition:
An **event** (denoted by any letter) is a **certain set of outcomes** of an experiment.

Example:

- Let C be an event in experiment E_5 defined as

$$C = \{\text{all outcomes consisting of an even number of heads}\}$$

Reasoning:
Events group outcomes based on a condition of interest and are the primary objects to which probabilities are assigned.

---

3. Sample Space

3.1 Definition

Sample space (S):
The sample space is the collection or set of **all possible distinct outcomes** of an experiment.

Key properties stated in the slides:

- **Mutually exclusive:** Only one outcome can occur at a time.
- **Collectively exhaustive:** No outcome outside the sample space is possible.

3.2 Types of Sample Spaces

The lecture classifies sample spaces as:

- **Discrete**
- **Countably infinite**
- **Continuous**

Examples listed in the slides:

- Flipping a fair coin once
- Rolling a cubical die with numbered faces
- Rolling two dice
- Flipping a coin until tails occurs
- Random number generator with interval $[0, 1)$

Logical importance:
The nature of the sample space determines how probability axioms and summations are applied.

---

4. Probability and Axioms of Probability

4.1 Definition of Probability

Probability is defined as:

- A measure of the likelihood of various events, or
- A function of an event that produces a numerical quantity measuring the likelihood of that event.

4.2 Probability Axioms

The axioms are taken to be **self-evident** and require no proof.

Let A be any event:

1. **Non-negativity and boundedness**

$$0 \leqslant \Pr(A) \leqslant 1$$

2. **Certain event**

$$\Pr(S) = 1$$

3. **Additivity for mutually exclusive events**
   - If $A \cap B = \varnothing$, then

$$\Pr(A \cup B) = \Pr(A) + \Pr(B)$$

4. **Countable additivity**
   - For mutually exclusive events A_i, $i = 1, 2, \ldots$,

$$\Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \Pr(A_i)$$

Logical role:

These axioms fully define the mathematical structure of probability.

---

5. Corollary from Probability Axioms

Corollary 2.1

For a **finite** number M of mutually exclusive events:

$$\Pr\left(\bigcup_{i=1}^{M} A_i\right) = \sum_{i=1}^{M} \Pr(A_i)$$

Explanation from slides:

- Axiom 3 reduces to this corollary when the sample space is finite.
- The general form of Axiom 3 is necessary when the sample space is infinite.

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6. Propositions from Probability Axioms

Proposition 2.1 (Complement Rule)

$$\Pr(A^c) = 1 - \Pr(A)$$

Step-by-step reasoning:

1. Event A and its complement A^c are mutually exclusive.
2. Their union is the sample space S.
3. Since $\Pr(S) = 1$, subtracting $\Pr(A)$ gives $\Pr(A^c)$.

Proposition 2.2 (Monotonicity)

If $A \subset B$, then:

$$\Pr(A) \leqslant \Pr(B)$$

Reasoning:

Event B contains all outcomes of A, and possibly additional ones, so its probability

cannot be smaller.

---

7. Joint Probability

7.1 Motivation

Joint probability is introduced to study the probability of **two events occurring simultaneously**, rather than individually.

7.2 Definition

The **joint probability** of events A and B is defined as:

$$\Pr(A \cap B)$$

This represents the probability that **both A and B occur together**.

7.3 Examples Mentioned in Slides

- **Card Deck Example**
- **Costume Party Example**

In each case:

1. Events are first defined individually.
2. Their intersection represents simultaneous occurrence.
3. Probability is assigned to the intersection.

---

8. Conditional Probability

8.1 Motivation

Conditional probability addresses situations where **additional information** is available, i.e., when one event is known to have occurred.

8.2 Definition

For $\Pr(B) > 0$, the **conditional probability** of A given B is:

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

8.3 Step-by-Step Logical Interpretation

1. Knowing that B has occurred restricts the sample space to B.
2. Within this restricted space, favorable outcomes are those in $A \cap B$.
3. Dividing by $\Pr(B)$ normalizes the probability within the reduced sample space.

8.4 Examples Listed in Slides

- Cards without replacement
- Game of Poker
- The Missing Key

Each example follows the same logical sequence:

1. Identify the conditioning event.
2. Restrict the sample space.
3. Apply the conditional probability formula.

---

9. Key Exam Takeaways

- Clearly distinguish between experiments, outcomes, events, and sample spaces.
- State axioms before using them in derivations.
- Joint probability corresponds to event intersections.
- Conditional probability is defined using joint probability and normalization.
- Always verify $\Pr(B) > 0$ when computing $\Pr(A \mid B)$.

End of Lecture 4 Scribe

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